Physics 12 Winter 2026
Week 7 Discussion

TA: Jiashu Han
February 17, 2026

1. **Summary of Topics from Last Week**

   - *From the lectures*
     - ◇ Nuclear fission, chain reaction
     - ◇ Nuclear reactor, moderator, coolant
       - ∗ Light-water reactor
       - ∗ Breeder reactor
       - ∗ Potential dangers, waste disposal
     - ◇ Nuclear binding energy
     - ◇ Nuclear fusion, confinement methods (inertial, magnetic)
     - ◇ Electromagnetism
       - ∗ Electric charge, Coulomb's law, Electric field, Electric current
       - ∗ Electric potential energy, Electric potential/voltage
       - ∗ Ohm's law, Electric circuit
       - ∗ Magnetic field, Magnetic materials, Permanent magnets, Curie temperature, Electromagnet
       - ∗ Lorentz force (on moving charge / current carrying wire)
       - ∗ Electromagnetic induction
       - ∗ Electric motor (including speakers) / generator
   - *From the textbook* (Wolfson 4th edition)
     - ◇ Nuclear reactors (LWR, breeder) (Chapter 7.5)
     - ◇ Uranium enrichment, nuclear waste (Chapter 7.6)
     - ◇ Environmental and health impacts (radiation dose) (Chapter 7.7-7.8)
     - ◇ Nuclear fusion (Chapter 7.9)

2. **Brain-warmer/mock quiz.**

(a) **Coulomb's Law.** Immediately after two separated charged particles are released from rest, both increase in speed. The sign of charge of the particles is therefore

A. same
B. opposite
C. either same or opposite; it does not matter
D. needs more info
E. particle speeds cannot change due to momentum conservation

Answer: C. In either case, each particle starts with zero velocity and feels a non-zero force that does not change sign over time, so it will keep accelerating in that direction, leading to increasing speed.

(b) **Nuclear reaction part 1.** Which of the following is true? Select all that apply.

A. Unlike in a chemical reaction, a nuclear reaction changes the structure of the nucleus and could potentially change an element into another.
B. Energy is released in a chemical reaction but not in a nuclear reaction.
C. Most chemical reactions change a molecule into different molecules, but the nuclei remain the same.
D. If a chemical reaction releases or absorbs energy, there is a mass difference between the left-hand side and right-hand side of the equation describing the reaction.

Answer: A, C and D

(c) **Nuclear reaction part 2.** Which of the following are examples of nuclear reactions? Select all that apply.

A. Alpha decay
B. Beta decay
C. Combustion
D. Sublimation
E. Nuclear fusion

Answer: A, B, and E.

(d) **Deuterium fusion.** In the homework this week, we considered the deuterium-deuterium fusion reaction:

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 3.27\ \mathrm{MeV} \tag{1} $$

What is the energy released per nucleon?

A. 1.635 MeV

B. 1.09 MeV

C. 2.18 MeV

D. 654 keV

E. 817.5 keV

Answer: E.

(e) **Magnetic force on moving charge.** A positive electric charge is moving upward in a uniform magnetic field that points to the right. Which of the following are true? Select all that apply.

A. The work done by the magnetic force is positive.

B. The work done by the magnetic force is zero.

C. The work done by the magnetic force is negatve.

D. The charge will turn into the page.

E. The charge will turn to the left.

F. The charge will turn out of the page.

G. The charge will turn to the right.

Answer: B and D.

(f) **Loudspeaker.** Which of the following are typical component of a loudspeaker? Select all that apply.

A. A permanent magnet.

B. Heating coil.

C. Cooling water.

D. Cyclone.

E. Electric-current-carrying wire

Answer: A and E.

3. **Uranium-235 fission, with more clarity**[1]**.** A typical example of an induced fission process in U-235 is:

$$\mathrm{n} + {}^{235}\mathrm{U} \rightarrow {}^{137}\mathrm{I} + {}^{96}\mathrm{Y} + 3\mathrm{n} \tag{2}$$

Calculate the actual fission energy released (in units of MeV) from this process using the following atomic masses:

$m({}^{235}\mathrm{U}) = 235.0439231$ u

[1]From last week's Riddle Me These worksheet

$m(^{137}\mathrm{I}) = 136.91787084$ u

$m(^{96}\mathrm{Y}) = 95.91589779$ u

Some additional numbers you may need: 1 u $= 1.66054 \times 10^{-27}$ kg; 1 J $= 6.242 \times 10^{12}$ MeV; $c = 2.99792 \times 10^{8}$ m/s.

Solutions: The excess energy is given by the energy equivalence of the difference between masses on the left-hand side of the equation and those on the right-hand side. We first find the mass difference:

$$\begin{aligned}\Delta m &= [m_{\mathrm{n}} + m(^{235}\mathrm{U})] - [m(^{137}\mathrm{I}) + m(^{96}\mathrm{Y}) + 3m_{\mathrm{n}}] \\ &= [(1.008664904\ \mathrm{u} + 235.0439231\ \mathrm{u}) - (136.91787084\ \mathrm{u} + 95.9191589779\ \mathrm{u} \\ &\quad +3 \times 1.008664904\ \mathrm{u})] \\ &= 0.1896\ \mathrm{u} \end{aligned} \tag{3}$$

There are two ways to proceed: We can either convert this to kg and use $\Delta E = \Delta m c^2$ to find the energy in J and then convert that to MeV, or find the energy in MeV corresponding to 1 u using $E = mc^2$ first and then multiply that directly to this number. Let's try the latter. The energy corresponding to 1 u is

$$\begin{aligned} E = mc^2 &= 1\ \mathrm{u} \times \frac{1.66054 \times 10^{-27}\ \mathrm{kg}}{1\ \mathrm{u}} \times (2.99792 \times 10^{8}\ \mathrm{m/s})^2 \times \frac{6.242 \times 10^{12}\ \mathrm{MeV}}{1\ \mathrm{J}} \\ &= 931\ \mathrm{MeV} \end{aligned} \tag{4}$$

Therefore, the energy difference corresponding to Δm is

$$\Delta E = 0.1896\ \mathrm{u} \times \frac{931\ \mathrm{MeV}}{1\ \mathrm{u}} \approx 180\ \mathrm{MeV} \tag{5}$$

4. **Energy per nucleon for U-235 fission.** What is the energy per nucleon released in the reaction in the previous problem?

   Solutions: The energy per nucleon released is the amount of total energy released divided by the number of nucleons that goes **into** the reaction. There are 1+235 = 236 nucleons that goes into this reaction

$$\frac{180\ \mathrm{MeV}}{236\ \mathrm{nucleons}} = 0.763\ \mathrm{MeV\ per\ nucleon} = 763\ \mathrm{keV\ per\ nucleon} \tag{6}$$